ETAS RTA-A2L 12.7.0

RTA-A2L Getting Started Guide

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ETAS RTA-A2L 12.7.0 - RTA-A2L Getting Started Guide R01 EN – 05.2025

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1 Introduction

This Getting Started Guide provides the user with the information required to install and use the RTA-A2L Generator.

1.1 Purpose

This document outlines the steps needed to install the RTA-A2L Generator and also offers advice on usage.

Intended Use

All ETAS RTA-CAR users who want to generate an A2L file for the AUTOSAR/ELF /ADX components they have created.

1.2 Document Conventions

`OCI_CANTxMessage msg0`	Code snippets are shown in a monospaced Inconsola typeface.
Choose **File→Open**.	Menu commands are shown in boldface.
Click **OK**.	Buttons are shown in boldface.
Press <ENTER>.	Keyboard commands are shown in angled brackets.
The 'Open File' dialog box is displayed.	Names of program windows, dialog boxes, fields, etc. are shown in quotation marks.
Select the file *setup.exe*	Text in drop-down lists on the screen, program code, as well as path- and file names are shown in italics.
A *distribution* is always a one-dimensional table of sample points.	General emphasis and new terms are set in italics.

1.3 Privacy Statement (GDPR)

Your privacy is important to ETAS so we have created the following Privacy Statement that informs you which data are processed in RTA-CAR, which data categories RTA-CAR uses, and which technical measures you have to take to ensure users' privacy.

Where relevant, we also provide additional information on where RTA-CAR stores personal or personal-related data, and how it can be deleted.

1.3.1 Data Processing

Personal or personal-related data or data categories are processed if using the ETAS License Manager (LiMa). The purchaser of this product is responsible for the legal conformity of processing the data in accordance with Article 4 No. 7 of the General Data Protection Regulation (GDPR). As the manufacturer, ETAS GmbH is not liable for any mishandling of this data.

When using the ETAS License Manager in combination with user-based licenses, the following personal or personal-related data or data categories are recorded

on the license server for the purposes of license management:

- Communication data: IP address
- User data: UserID, WindowsUserID

1.3.2 Technical and organizational measures

This product does not encrypt the personal or personal-related data or data categories that it records. Ensure that the data recorded are secured by means of suitable technical or organizational measures in your IT system. Personal or personal-related data in log files can be deleted by tools in the operating system.

1.4 Safety Notice

For any safety-related development that uses the RTA-CAR tool you should apply an applicable safety standard such as ISO 26262:2018.

The ETAS_Safety_Advice.pdf provides additional safety information.

2 System Requirements

Please refer to the latest RTA-CAR Release Notes for more details.

3 Installation

3.1 Overview

There are two key components to RTA-A2L:

- The RTA-A2L software, plugins and documentation.
- The ETAS License Manager (LiMa) available through RTA-CAR.

RTA-A2L does not write to the registry nor require local admin rights. It can be copied from one location or machine to another; the software relies on the relative locations of files and subdirectories under the installation folder in order to function.

RTA-A2L is bundled with RTA-CAR. If a new RTA-A2L version is required, RTA-CAR has to be upgraded.

3.2 Licensing

3.2.1 ETAS License Manager

The RTA-CAR installer can optionally install the ETAS License Manager.

- If you already have LiMa installed (e.g. you have other ETAS software installed that uses it) then you do not have to re-install it.
- If your organization has already made arrangements for licensing ETAS products as part of a centralized rollout scheme, then you may not need to install LiMa.
- If you do not have local admin rights, then you cannot install LiMa.

3.2.2 Keys

RTA-A2L is protected by license keys installed using the ETAS License Manager. If you require a license, please request one using the ETAS License Manager or by contacting ETAS GmbH.

License keys for RTA-A2L are normally locked to the MAC address of the Ethernet controller in your host PC. Other licensing options are available. Please contact ETAS to find out more.

Node-locked licenses are supplied in an ASCII text file, which will be sent to you upon completion of a valid license agreement. Until a valid license is installed, the ETAS License Manager will display the RTA-A2L License State as “NOT AVAILABLE” and the backend processors will not run.

The license file is installed onto your host PC using the ETAS License Manager. This can be performed by selecting **File→Add License File** from the menu:

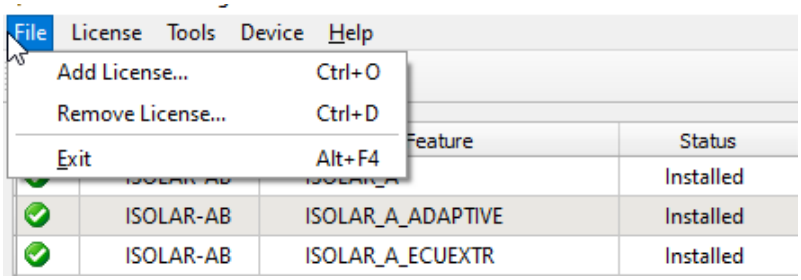
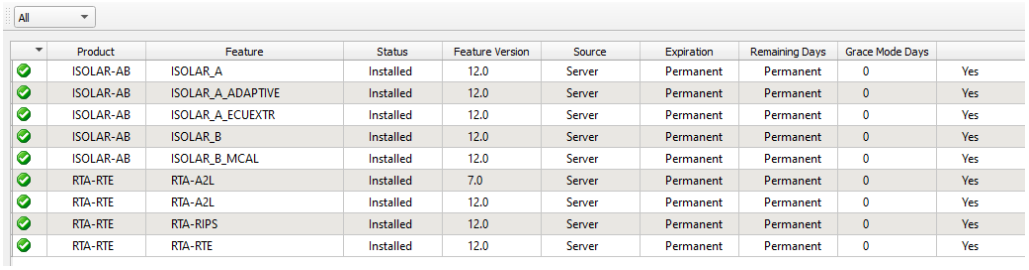


Figure 3.1.Add License File to ETAS License Manager

When a valid license has been installed, the ETAS License Manager will display the license version, status, expiration date and source:



Product	Feature	Status	Feature Version	Source	Expiration	Remaining Days	Grace Mode Days	
ISOLAR-AB	ISOLAR_A	Installed	12.0	Server	Permanent	Permanent	0	Yes
ISOLAR-AB	ISOLAR_A_ADAPTIVE	Installed	12.0	Server	Permanent	Permanent	0	Yes
ISOLAR-AB	ISOLAR_A_ECUEXTR	Installed	12.0	Server	Permanent	Permanent	0	Yes
ISOLAR-AB	ISOLAR_B	Installed	12.0	Server	Permanent	Permanent	0	Yes
ISOLAR-AB	ISOLAR_B_MCAL	Installed	12.0	Server	Permanent	Permanent	0	Yes
RTA-RTE	RTA-A2L	Installed	7.0	Server	Permanent	Permanent	0	Yes
RTA-RTE	RTA-A2L	Installed	12.0	Server	Permanent	Permanent	0	Yes
RTA-RTE	RTA-RIPS	Installed	12.0	Server	Permanent	Permanent	0	Yes
RTA-RTE	RTA-RTE	Installed	12.0	Server	Permanent	Permanent	0	Yes

Figure 3.2.Valid installation of license feature

3.2.3 Verifying the License

RTA-A2L will attempt to check out a license. If it succeeds, it will display the License Feature as validated.

3.3 Installation Contents

Let's say the RTA-A2L application is located at *ApplicationsRTA-A2L*. The folder contains:

- The source code for the software components. These files have extension *.java*.
- An Eclipse executable to run RTA-A2L and generate the A2L file.

3.3.1 Invoke A2L Generation for Windows

Assuming that RTA-A2L has been unzipped to a Windows directory location, the Eclipse executable, *RTA-A2L.exe* runs at the command line with options in the file *rta-a2lOptions.txt*:

```
RTA-A2L -vm <java21.exe> --optionsfile <optionsfile>
```

NOTE: If RTA-A2L is delivered together with RTA-CAR, java package can be found in ISOLAR-AB folder.

The tool is executed by passing the JVM path and the optionsfile (and its path):

```
C:\tool\RTA-A2L_Standalone\RTA-A2L_12.7.0\eclipse\RTA-A2L.exe -vm
 <ISOLAR-ABFolder>\javalibs\jdk\bin\java.exe -optionsfile C:\tool\Test\RTA-A2L\a2loptionfile.opt
```

3.3.2 Invoke A2L Generation for Linux

Assuming that RTA-A2L has been unzipped to a Linux directory location, the Eclipse executable, *RTA-A2L* runs at the command line with options in the file *rta-a2lOptions.txt*:

```
RTA-A2L -vm <java21.exe> --optionsfile <optionsfile>
```

(*RTA-A2L* is the Linux executable.)

The tool is executed by passing the JVM path and the optionsfile (and its path):

```
/home/RTA-A2L_Standalone/RTA-A2L_12.3.0/eclipse/RTA-A2L -vm <ISOLAR-ABFolder>/javalibs/jdk/bin/java.exe
-optionsfile home/Test/RTA-A2L/a2loptionfile.opt
```

Building an Example Application

Please see the RTA-A2L Reference Manual for full documentation of available command line options.

3.3.3 Generated Files

The RTA-A2L Generator creates the following files:

File	Description
RTA-A2L.a2l	The AUTOSAR A2L file is generated by processing the *Rte_Mc-SupportData.arxml* file.
RTA-A2L_NonAR.a2l	The Non-AUTOSAR A2L file is generated by processing the ADX file specification.
RTA-A2L.adx	XML format with mapping information of A2L names and C label names for the same labels.
RTA-A2L_tool.txt	Tool log file. Logs all the warnings and errors encountered during the generation of the A2L file.

4 Documentation

RTA-A2L includes the following manuals:

- RTA-A2L Getting Started Guide i.e. this manual.
- RTA-A2L Release Notes i.e. details of changes to RTA-A2L by version.
- RTA-A2L Reference Manual i.e. the guide that describes in detail the underlying principles of RTA-A2L and its features.
- RTA-A2L Error Reference i.e a list of error messages plus hints on how to fix them.

5 ETAS Contact Addresses

5.1 Technical Support

Technical support is available to all users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office.

- https://www.etas.com/en/contact.php

The best way to get technical support is using the online support portal RTA Hotline:

- https://rtahotline.etas.com/

It is helpful if you can provide technical support with the following information (where relevant):

- Your support contract number
- The version of the ETAS tools you are using
- The version of the compiler tool chain you are using
- The command line (or reproduction of steps) that result in an error message
- The error messages or return codes you received (if any)
- Your RTA-CAR Project or if not possible the arxml file(s) containing the error(s).
- The *Diagnostic.dmp* if it was generated

5.2 ETAS HQ

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